COUNTRY PROFILES

Australia's Uranium

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- Australia's uranium has been mined since 1954, and three mines are currently operating. More are planned.
- Australia's known uranium resources are the world's largest – almost one-third of the world total.
- In 2022 Australia produced 4820 tonnes of U_3O_8 (4087 tU). In 2022 it was the world's fourth ranking producer, producing 8% of global uranium.
- All Australia's uranium production is exported.

Privacy - Terms

- Australia uses no nuclear power, but with high reliance on coal any likely carbon constraints on electricity generation will make it a strong possibility.

- In May 2016 the South Australian government's royal commission on the nuclear fuel cycle reported. Its main recommendation was for an international high-level nuclear waste repository, though this was not accepted.

The Australian economy is unique in the OECD in that the mining sector makes up the largest sector share of national GDP – 10.4% between 2019 and 2020. Uranium is a small part of this economically, but is significant in energy terms.

- In the 1930s ores were mined at Radium Hill and Mount Painter in SA to recover radium for medical purposes. As a result a few hundred kilograms of uranium were also produced.

Uranium ores as such were mined and treated in Australia initially from the 1950s until 1971. Radium Hill, SA, Rum Jungle, NT, and Mary Kathleen, Queensland, were the largest producers of uranium (as yellowcake). Production ceased either when ore reserves were exhausted or contracts were filled. Sales were to supply material primarily intended for USA and UK weapons programs at that time. However, much of it was used for electricity production.

The development of civil nuclear power stimulated a second wave of exploration activity in the late 1960s. A total of some 60 uranium deposits were identified from the 1950s through to the late 1970s, many by big companies with big budgets. (Since then only two significant new ones have been found: Kintyre and Beverley Four Mile. The minor exploration boom 2002-07 was driven by small companies focused on proving up known deposits.)

Mary Kathleen began recommissioning its mine and mill in 1974. Other developments were deferred pending the findings of the Ranger Uranium Environmental Inquiry, and its decision in the light of these. Mary Kathleen's second production phase was1976 to the end of 1982.

The Commonwealth Government announced in 1977 that new uranium mining was to proceed, commencing with the Ranger project in the Northern Territory. This mine opened in 1981. In 1979, Queensland Mines opened Nabarlek in the same region of Northern Territory. The orebody was mined out in one dry season and the ore

stockpiled for treatment from 1980. The mine site is now rehabilitated.

A brief history of Australian uranium mining is appended. See also Former Australian Uranium Mines appendix.

Australian Uranium Production and Exports

	Calendar year	2009	2010	2011	2012	2013	2014	2015	2016	201
Production	tonnes U_3O_8	9413	6958	7056	8244	7488	5897	6668	7447	693
	tonnes U	7982	5900	5983	6991	6350	5000	5654	6315	588
Exports	tonnes U_3O_8	9706	6888	6628	8116	7317	5669	6969	7679	675
	tonnes U	8230	5841	6170	6882	6205	4807	5909	6511	572
	A$ million FOB	1116	608	586	776	704	504	802	715	528
Export value*	A$/kg U_3O_8	114.9	88.3	88.4	95.6	96.2	88.9	115.1	93.1	78.:

Source: Companies, DRET, DIIS
* average $A from declared net FOB estimates

Recent Production from Australian Uranium Mines (tonnes of U_3O_8)

	2015	2016	2017	2018	2019	2020	2021	2022
Olympic Dam	3728	3813	2808	3736	3967	3611	1922	3317
Ranger	2005	2351	2294	1999	1751	1574	29	0
Beverley	0	100	20	0	0	0	0	0
Four Mile	935	1183	1815	1961	2080	2130	2241	1503
Honeymoon	0	0	0	0	0	0	0	0
Total	6668	7447	6937	7696	7798	7315	4192	4820

Operating Mines

The **Ranger mine** and associated town of Jabiru is about 230 kilometres east of Darwin, in the Northern Territory, surrounded by the Kakadu National Park, a major tourist attraction. The mine opened in 1981 at a production rate of approximately 3300 tonnes per year of uranium oxide and has since been expanded to 5500 t/yr capacity. Mining of the second pit was 1997 to 2012, and this is now being backfilled. Treatment is conventional acid leach. Any future development will be underground, and application was made for approval of this in January 2013. Substantial development was undertaken to mine the Ranger Deeps, and in June 2015 ERA announced that it will defer proceeding further with development of the underground mine to access 27,650 tonnes of uranium, after spending A$ 177 million on the project. This is due both to slow recovery in the uranium market and the requirement to cease operations under the present Ranger Authority, which expired in January 2021. Negotiations are exploring the potential to extend the deadline. Ranger is owned by Energy Resources of Australia Ltd (ERA), for many years a 68.39% subsidiary of Rio Tinto, but becoming 86.33% owned in February 2020 to ensure that it had funds for rehabilitation. Production ceased in January 2021.

During 1988 the **Olympic Dam** project, then a joint venture of Western Mining Corporation and BP Minerals, commenced operations about 560 km north of Adelaide, in an arid part of South Australia. The massive deposit is underground, some 350 metres below the surface, and is the largest known uranium orebody in the world. The large underground mine produces copper, with gold and uranium as major by-products. Annual production capacity for uranium oxide has been expanded from 1800 to 4600 tonnes U_3O_8. It is now owned by BHP Billiton, following its 2005 takeover of WMC Resources. There were plans to greatly increase the mine's size and output, by accessing the orebody with a huge open pit, about 4.1 x 3.5 km and 1000m deep,but since 2015 only underground development is planned. (Further details below.)

About 80% of the uranium is recovered in conventional acid leach of the flotation tailings from copper recovery. Most of the remaining 20% is from acid leach of the copper concentrate, but that concentrate then still contains up to 0.15% uranium. Hence the copper must be smelted at site, since selling it to overseas smelters would create both processing and safeguards complications for the smelter operator. This could change as part of a major envisaged expansion.

Both Ranger and the now-closed and rehabilitated Nabarlek mines are on aboriginal land in the Alligator Rivers region of the Northern Territory. Aboriginal people receive royalties of 4.25% on sales of

uranium from Northern Territory mines. The total received simply from Ranger is now over $207 million, and $14 million came from Nabarlek.

Energy Resources of Australia (ERA) expects to complete its 2022 feasibility study on the rehabilitation of the Ranger Project Area in September 2023.

The Olympic Dam mine is on formerly pastoral land in the middle of South Australia. A town to accommodate 3500 people was built at Roxby Downs to service the mine. The 18,000 ha mine lease is managed as a nature reserve.

The **Beverley** mine in South Australia started operation late in 2000, 520 kilometers north of Adelaide, on the plains north-west of Lake Frome. It was Australia's first in situ recovery (ISR) mine, accessing a palaeochannel deposit in sand in a saline aquifer. It was licensed to produce 1180 t/yr U_3O_8 (1000 tU), and reached this level in 2004. It is owned and operated by Heathgate Resources Pty Ltd, an affiliate of General Atomics (GA). In December 2010, the company received government approval to mine the Beverley North deposits. Mining of Beverley ceased at the end of 2013, and of Beverley North early in 2014, though 100 tonnes was reported for the second half of 2016.

In 2014, Heathgate began ISR mining operations on the contiguous **Four Mile** deposits, owned by GA affiliate Quasar Resources. Four Mile resources are split across west, east, and northeast orebodies. Uranium processing continues at the Beverley processing plant with pregnant lixiviant pumped directly from the Four Mile wellfields. Production remains steady at about 2000 t U_3O_8 per year.

In 2015, Quasar purchased Alliance Resources' 25% interest in the Four Mile uranium project, including its share of uranium oxide concentrates already mined. The project is now wholly owned.

The **Honeymoon** ISL mine in South Australia commenced operation in 2011. The owners received government approval to proceed with ISL mine development in November 2001 but reassessed its ore reserves and Uranium One, based in Toronto, finally moved to development in 2007. In 2008 Mitsui agreed to join the project as 49% joint venture partner, and a construction contract was then let. Operations were ramping up to 400 t/yr. In 2012 production was expected to be 275 tonnes U_3O_8, at $47/lb - three times the average cost of production in Kazakhstan. In fact it produced less. Mitsui largely funded the development and commissioning, but then withdrew from the project in 2012. In November 2013 Uranium

One, by then owned by Russia's ARMZ, closed the mine and put it on care and maintenance until uranium prices improved. In September 2015 Boss Resources Ltd, now Boss Energy Ltd, based in Perth agreed to buy Uranium One Australia which owned the mine. Boss completed a definitive feasibility study in January 2020 which provided a base case to rapidly restart uranium production at 907 t U_3O_8 (769 tU) per year over a 12-year life-of-mine. In June 2022 Boss approved the final investment decision for the development of the in situ leach project and said production was expected to begin during the final quarter of 2023.

For more details of mines see Appendix: Australia's Uranium Mines.

Uranium resources

On the basis of 2019 data Australia has 28% of the world's uranium resources (under US$ 130/kg) – 1.7 million tonnes of uranium. Almost half of Australia's reasonably assured resources of uranium in this price category was actually in the under $80/kg U category when this was last reported. The vast majority of Australia's uranium resources (to $130/kgU) are within five deposits: Olympic Dam (the world's largest known uranium deposit), Ranger, Jabiluka, Kintyre and Yeelirrie.

The world's reasonably assured plus inferred resources in the $130/kg category are tabulated in the Supply of

Uranium information page.

A review of Australia's uranium is provided on the Geoscience Australia website.

Uranium resources at mines and major deposits
tonnes U_3O_8

Mine or deposit	Type	Reserves	Measured & Indicated Resources	Inferred Resources
Ranger	hard rock, most underground	8,081	44,883	11,087
Olympic Dam	hard rock, underground	305,150	1,555,100	1,012,800
Beverley	palaeochannel, ISL		?	?
Four Mile	palaeochannel, ISL		14,520	39,926
Honeymoon	palaeochannel, ISL		4,920	7,620
Jabiluka	hard tock, underground		82,945	54,162
Kintyre	hard rock		25,274	2,400
Yeelirrie	calcrete		57,760	0
Wiluna	calcrete		27,400	9,600
Mulga Rock	palaeochannel and lignite		20,650	20,000
Samphire	palaeochannel and basement granite		0	21,000
Valhalla	hard rock		24,765	5,860

Resources are additional to reserves.

Prospective mines and expansion

The Jabiluka uranium deposit in the Northern Territory was discovered in 1971-73, 20 kilometres north of Ranger. It is surrounded by the Kakadu National Park, but the mine lease area is excluded from the National Park and adjoins the Ranger lease. It has resources of over 130 000 tonnes of uranium oxide, and is one of the world's larger high-grade uranium deposits. A mining lease was granted in 1982. Then with the Australian Labor Party coming to power in the 1983 federal election, Commonwealth approval was withdrawn and development ceased. In 1991 Energy

Resources of Australia (ERA), the operator of the adjacent Ranger mine, bought the Jabiluka lease from Pancontinental for A $125 million.

Following the 1996 change of government and further approvals, development of the underground mine proceeded with an 1150 metre access decline and a further 700 metres of excavation around the orebody. However, mining was deferred until agreement could be reached regarding treatment of Jabiluka ore at the Ranger mill. ERA (whose parent company is Rio Tinto) will not proceed with the mine until there is agreement from the local Mirrar Aboriginal people.

Cameco holds or has applied for exploration tenements on almost one million hectares of prospective land in the western part of Arnhem Land, Northern Territory. In 2012 and 2013 Cameco and CSIRO conducted a collaborative research study exploring the effectiveness of termitaria (termite mounds) sampling in detecting uranium and other pathfinder elements. The findings of the study indicated that the presence of uranium in termite mounds is most likely due to "the consumption of vegetation that has drawn soluble uranium, produced during the weathering of a source, up from the water table." Termite orientation studies have returned encouraging results, however the study found that termitaria techniques were "more effective at providing a vector to sub-surface uranium mineralisation."

For Olympic Dam BHP Billiton undertook a major feasibility study on greatly expanding the mine, and in 2009 it released the 4600-page environmental impact statement for the project. This was approved by state and federal governments in October 2011. The plan was to develop a large open pit with associated infrastructure over 11 years and lift uranium production to 19,000 tonnes U_3O_8 per year. The open pit would mean that up to 98% of the ore is mined rather than much less of it. Most of the uranium would be separated at the mine, but up to 2000 t/yr would be exported in copper concentrates, requiring a smelter for these in China or Japan which is subject to international nuclear safeguards. New infrastructure would include a 280 ML/day desalination plant on Spencer Gulf, supplying 200 ML/day to the operation, and 650 MWe increase in power supply. The present underground mining would continue in the narrow northern part of the orebody. However, in August 2012 the company said that it would investigate a new and less-costly design for its planned open-pit expansion, which meant it could not approve the project in time to meet a government deadline in December. In November 2012 the state government granted a four-

year extension, conditional on the company spending $650 million on pre-project research on heap leaching and on community work.

In November 2014, in a general announcement about productivity, BHP Billiton flagged a 27% increase in copper production at Olympic Dam from 2018, and a doubling from that level subsequently by "a low-risk underground expansion with significantly lower capital intensity than the previous open cut design. This has the potential to deliver over 450,000 tonnes of copper production a year at first quartile C1 costs by the middle of next decade". The uranium implications were not mentioned, but assuming the same ore as today, it would mean about 5000 t U_3O_8 (4200 tU) per year from 2018 and some 9400 t U_3O_8 (8000 tU) per year in mid-2020s. In July 2016 the company confirmed that it would now focus on underground development only and not pursue the earlier open pit plan. Its increased copper production would involve a corresponding increase in uranium production.

In late October 2022 a planned expansion of Olympic Dam was scrapped.

Cameco and Mitsubishi (70:30%) bought the **Kintyre** deposit in WA in 2008 from Rio Tinto for uS$ 495 million. Cameco initially envisaged starting mine construction in 2013 and operation in 2015, to produce 2700 to 3600 t U_3O_8 per year for 15 years. In mid-2012 Cameco put the project on hold pending firmer uranium prices or lower development costs. State and federal environmental approvals were given in 2015.

BHP Billiton applied to bring its **Yeelirri**e, WA, deposit into production and projected 2000 t/yr U_3O_8 production from 2014, though in February 2010 approval was sought for production at 3500 t/yr. However, in 2011 the project was wound down due to high treatment costs and in 2012 it was sold to Cameco for US$ 340 million. In November 2014 Cameco requested the WA EPA to cancel the earlier environmental application, and submit a new one involving production at 7500 t U_3O_8 per year, and to assess the application under new 2012 EPA procedures. In January 2017 the state government approved development.

Toro Energy is well advanced with plans to produce 900 t/yr U_3O_8 from its **Wiluna** project, comprising the shallow Lake Way and Centipede-Millipede deposits and the nearby Lake Maitland deposit in WA, from 2016. In January 2017 the state government approved development, but the company awaits higher uranium prices.

Vimy Resources is developing the **Mulga Rock** deposits in WA to produce 1300 t/yr U_3O_8, with progressive open pits to 32 metres depth. In December 2016 the state government approved development and federal government approval followed in March 2017. In December 2021 Vimy Resources announced that excavation had begun on the Ambassador North deposit at Mulga Rock.

The largest prospective Queensland mine is Paladin's **Valhalla**, 40 km north of Mount Isa. This is a major deposit but was stalled to 2012 by state government policies.

There has been increasing foreign equity in Australian uranium deposits. As well as the Honeymoon, Kintyre and Yeelirrie projects above, in February 2009 Mega Uranium sold 35% of the Lake Maitland project to the Itochu Corporation (10% of Japanese share) and Japan Australia Uranium Resources Development Co. Ltd. (JAURD), acting on behalf of Kansai Electric Power Company (50%), Kyushu Electric Power Company (25%) and Shikoku Electric Power Company (15%) for US$ 49 million. In 2006 Sinosteel bought 60% of Pepinini's Curnamona project for A$ 31 million, and in 2009 China Guangdong NPC bought 70% of Energy Metals' Bigrlyi project for A$ 83.6 million. Both are early-stage exploration ventures.

For more details of mine prospects see Appendix on Australia's Uranium Deposits and Prospective Mines.

Despite restrictive state government policies and perhaps in anticipation of their disappearance, uranium exploration gathered pace during 2006, with more than 200 companies professing an interest, compared with 34 the previous year, and A$ 80 million being spent. Expenditure then more than doubled, to A$ 182 million in 2007, A$ 227 million in 2008, A$ 180 million in 2009 and A$ 190 million in 2011. It then declined abruptly to A$ 98 million in 2012.

Uranium exploration has been illegal in Victoria and New South Wales, and remains so in Victoria. Uranium mining is being reinstated in Queensland after a few years' break.

Economic benefits of mining uranium

About 1400 people are employed in uranium mining, at least 500 in uranium exploration, and about 60 jobs are in regulation of uranium mining.

Uranium exports

Australian production is all exported. In 2022 Australia produced 4087 tU of uranium, 8% of global production. Uranium comprises about 17% of the country's energy exports in thermal terms.

Australia's uranium is sold strictly for electrical power generation only, and safeguards are in place to ensure this. Australia is a party to the Nuclear Non-Proliferation Treaty (NPT) as a non-nuclear weapons state. Its safeguards agreement under the NPT came into force in 1974 and it was the first country in the world to bring into force the Additional Protocol in relation to this – in 1997. In addition to these international arrangements Australia requires customer countries to have entered a bilateral safeguards treaty which is more rigorous than NPT arrangements.

The value of Australia's uranium oxide concentrate exports is considerable, and in 2021 it was over A$ 700 million.

An indication of the nations which currently purchase Australia's uranium are set out below, though up-to-date details on country destinations are not available. All have a large commitment to nuclear power.

The USA generates around 30% of the world's nuclear power. Much of its 21,200 tonnes U_3O_8e of imported (2021) uranium comes from Canada, but Australia is a major source supplying almost 15% in 2021, more than half of total Australian exports. Europe depends heavily on nuclear power and EU countries are also major customers. In 2015, 1910 tU or 12% of EU uranium was from Australia. In 2020 13% (1671 t) of EU uranium was imported from Australia. Japan, South Korea, China and India are important customers due to their dependence on nuclear energy.

Customer countries' contracted imports of Australian uranium oxide concentrate – U_3O_8 – may be summarized as follows, though detailed information has not been readily available in recent years (see also the reactor table):

- USA: up to 5000 tonnes per year.

- EU: up to 3500 tonnes per year, including Belgium, Finland, France, Germany, Spain, Sweden, UK.
- Japan: formerly up to 2500 tonnes per year.
- South Korea: up to 1500 tonnes per year.
- China: about 500 tonnes per year.
- Taiwan: up to 500 tonnes per year.
- India: up to 300 tonnes per year likely from 2018.

Australia is a preferred uranium supplier to world, especially East Asian, markets where demand is growing most rapidly. In 2006 a bilateral safeguards agreement was concluded with China, enabling exports there, and in 2007 a similar agreement was signed with Russia, which came into force in 2010. An agreement with the United Arab Emirates (UAE) came into force in 2014, and another with India came into force in November 2015, at which time administrative arrangements for each were also finalized and they became operational. In 2016 a bilateral agreement was signed with Ukraine. This made the total 25 treaties covering 43 countries plus Taiwan. Australia could readily increase its share of the world market because of its low cost resources and its political and economic stability.

As well as uranium sold to overseas customers (mainly utilities) by the four mining companies, Energy Metals Ltd, an exploration company with majority Chinese ownership, has an export permit. In December 2011 it announced the sale of 68 tonnes of U_3O_8 to its parent company, China General Nuclear Power, for shipment in 2012. (CGN's wholly-owned subsidiary, China Uranium Development Co., is Energy Metals' largest shareholder with a 60.6% stake. The company sources uranium from Australian producers.)

Environmental aspects of uranium exports are notable: shipping approximately 6000 tonnes of U_3O_8 as in 2021 is the energy equivalent of shipping 120 million tonnes of thermal coal. Australia's present thermal coal exports are about 200 million tonnes, requiring between 6000 and 8000 voyages of bulk carriers through environmentally sensitive regions such as the Great Barrier Reef. Export coal also has an environmental impact through the provision of harbours and railways.

Nuclear power prospects

Australia uses no nuclear power, but with high reliance on coal any likely carbon constraints on electricity generation will make it a strong possibility.

Australia has a significant infrastructure to support any future nuclear power program. As well as the Australian Nuclear Science & Technology Organisation (ANSTO), which owns and runs the modern 20 MWt Opal research reactor, there is a world-ranking safeguards set-up – the Australian Safeguards & Non-proliferation Office (ASNO), the Australian Radiation Protection and Nuclear Safety Agency (ARPANSA) and a well-developed uranium mining industry.

However, in contrast to most G20 countries, the only real driver for nuclear power in Australia is reduction of CO_2 emissions, or costs arising from that. Apart from that, Australia's huge coal resources and significant natural gas underwrite energy security and provide low-cost power. (Many other countries have cost of electricity and energy security as major factors.)

There are several legal hurdles impeding consideration of nuclear power for Australia. NSW has a Uranium Mining and Nuclear Facilities (Prohibition) Act 1986, and Victoria has a Nuclear Activities (Prohibitions) Act 1983. Federally, the Environment Protection and Biodiversity Conservation Act 1999 and Australian Radiation Protection and Nuclear Safety Act 1988 will need to be amended to remove prohibitions against effective regulation of nuclear power.*

* The ARPANS Act 1998 has section 10 *Prohibition on certain nuclear installations*, and the EPBC Act 1999 much the same in section 140A. ARPANS Act section 10:

1. Nothing in this Act is to be taken to authorize the construction or operation of any of the following nuclear installations:

(a) a nuclear fuel fabrication plant;

- (b) a nuclear power plant;
- (c) an enrichment plant;
- (d) a reprocessing facility.

1. The CEO must not issue a licence under section 32 in respect of any of the facilities mentioned in subsection (1).

Fuller details and context are on the Bright New World website.

In November 2018 the International Atomic Energy Agency (IAEA) completed a 12-day integrated regulatory review service mission focused on ARPANSA to assess the regulatory framework for nuclear and radiation safety in Australia.

UMPNER report 2006 and follow-on

In December 2006 the report of the Prime Minster's expert taskforce considering nuclear power was released. It concluded that “the Review sees nuclear power as a practical option for part of Australia’s electricity production” and said nuclear power would be 20-50% more expensive than coal-fired power and (with renewables) it would only be competitive if "low to moderate" costs are imposed on carbon emissions (A$ 15-40 – US$ 12-30 – per tonne CO2). "Nuclear power is the least-cost low-emission technology that can provide base-load power" and has low life cycle impacts environmentally.

The then Prime Minister said that in the context of meeting increased energy needs while reducing greenhouse gas emissions "if we are to have a sensible response we have to include nuclear power". "The report provides a thorough examination of all aspects of the nuclear fuel cycle and the possible role of nuclear power in generating electricity in Australia in the longer term. It provides a clear and comprehensive analysis of the facts surrounding the nuclear industry and debunks a number of myths. I am certain that the report will make a significant contribution to informing public debate on these issues."

The report said that the first nuclear plants could be running in 15 years, and looking beyond that, 25 reactors at coastal sites might be supplying one-third of Australia's (doubled) electricity demand by 2050. Certainly "the challenge to contain and reduce greenhouse gas emissions would be considerably eased by

investment in nuclear plants." "Emission reductions from nuclear power could reach 8 to 18% of national emissions in 2050".

In April 2007 the Prime Minister announced that the government would proceed to open the way for nuclear power in Australia by setting up a nuclear regulatory regime and removing any regulatory obstacles which might unreasonably stand in the way of building nuclear power plants. Australia would also apply to join the Generation IV International Forum, which is developing advanced reactor designs for deployment about 2025. The government would also take steps to remove impediments to uranium mining. "Policies or political platforms that seek to constrain the development of a safe and reliable Australian uranium industry – and which rule out the possibility of climate-friendly nuclear energy – are not really serious about addressing climate change in a practical way that does not strangle the Australian economy."

In June 2007 the emissions trading taskforce report proposed that Australia should move steadily to implement an emissions trading scheme by 2012. While Australia cannot afford to wait upon a global regime, its own should be devised so as to avoid the shortcomings of present schemes and also articulate internationally. Both emission reduction targets and carbon price would be low initially and ramp up. The need for a trading scheme "more comprehensive, more rigorously grounded in economics and with better governance than anything in Europe" was noted. It would be designed to appeal to developing nations. The cost increment on coal-fired power generation brought about by a carbon emission cost would be likely to make nuclear power competitive in Australia.

With a change of government late in 2007 the move towards nuclear power was halted and the implementation of an emissions trading scheme became bogged down in political rhetoric.

National Generators Forum 2006

Any proposal for building nuclear power plants would need to be brought forward by generating companies. The National Generators Forum published a report in 2006 on *Reducing Greenhouse Gas Emissions from Power Generation* which concluded that "Stabilising emissions at present levels and meeting base-load requirements could be achieved with nuclear power at comparatively modest cost." While projected cost increases to 2050 could be more than 120%, using nuclear power would halve

the increase. "At $20 per tonne of CO2 price, nuclear starts to become more cost-effective than current fossil fuel technologies."

Cooling will be a major issue in respect to future base-load generating capacity in Australia. In 2020 about 55% of electricity was produced from coal-fired plants, mostly cooled by evaporating water in cooling towers. An estimated 300 GL/yr of fresh water is thus evaporated and lost – about the same as Melbourne's water use. In the light of widespread shortage of fresh water, cooling of nuclear plants would need to be by seawater, hence coastal sites would be required* and to the extent that nuclear plants actually replaced coal-fired plants, a very large amount of fresh water would be freed up for other uses. Coastal location of nuclear plants would also give rise to the possibility of cogeneration, using waste heat or surplus heat for desalination and production of potable water.

* A coal plant is normally sited on a coalfield (inland), so does use a lot of water for evaporative cooling towers. A nuclear plant can be anywhere, from the point of view of fuel supply.

IFNEC & GIF participation

In September 2007 Australia was one of eleven countries joining the five founders in the Global Nuclear Energy Partnership (now the International Framework for Nuclear Energy Cooperation – IFNEC). Australia made it a condition that it is not obliged to accept any foreign nuclear waste, and it reserved the right to enrich uranium in the future. In the lead up to this Australia and the USA finalized a joint action plan for civil nuclear energy cooperation including R&D and regulatory issues.

In April 2016 Australia became the 14th member of the Generation IV International Forum (GIF) which is working on advanced nuclear technologies, focused on six power reactor designs expected to be commercially viable from about 2030. ANSTO will be the means of contributing to GIF's goals. GIF was set up in 2001, and the technical secretariat is with the OECD's Nuclear Energy Agency in Paris, alongside IFNEC and another major international programme. The membership was formalized in September 2017.

ATSE action plan 2014

In July 2013 the Australian Academy of Technological Sciences and Engineering (ATSE) held a two-day conference on nuclear power for Australia. This resulted in an action plan in 2014, which included

comparing the nuclear option with alternative energy scenarios in the 2030-2050 timeframe, reviewing current policies which preclude consideration of nuclear power, commence open and active community engagement, building on overseas R&D program linkages.

SA royal commission 2015-16

In February 2015 the Labor state government of South Australia set up a royal commission into the potential for nuclear power in that state, which already produces two-thirds of Australia's uranium. The terms of reference included fuel cycle and high-level waste disposal as well as power generation. The inquiry was supported by the state Liberal (conservative) opposition and the federal Liberal coalition government, but not by the federal Labor party (though it supports uranium mining). Though the royal commission's findings were positive in many respects, several would require federal laws to be changed.

Insofar as the royal commission will direct future power investment in SA, the question of reactor unit size arises. At present the unit size of any generating unit there is regulated at 260 MWe, though modelling has shown 500 MWe units are possible. Small modular reactors would therefore be indicated. But if transmission links were expanded a SA nuclear power plant with large reactors could serve the eastern states.

In May 2016 the Royal Commission into the Nuclear Fuel Cycle reported. It found that it would not be commercially viable to develop a nuclear power plant in the state under current market rules, but noted that as "a low-carbon energy source comparable with other renewable technologies", nuclear may be required in the future. It therefore recommended that the South Australian government should "pursue removal at the federal level of existing prohibitions on nuclear power generation to allow it to contribute to a low-carbon electricity system, if required." It also called for the removal at the federal level of prohibitions on the licensing of fuel cycle facilities, although it noted that in a currently oversupplied market the provision of such services would not be commercially viable in the next decade.

In November the South Australian government said it would support nine of the 12 recommendations of its royal commission. Five accepted recommendations relate to uranium mining and exploration, and include the expansion of uranium mining. The government also supported the commission's recommendations to

collaborate at the national level “on the development of a comprehensive national energy policy that enables all technologies, including nuclear, to contribute to a reliable low-carbon electricity network at the lowest possible system cost,” and to monitor the development of new nuclear reactor designs. It also supported a recommendation to promote and increase the use of nuclear medicine at the South Australian Health and Medical Research Institute. It did not support the commission's recommendation that it should pursue the removal of existing prohibitions on nuclear power generation in the state, "recognising that in the short-to-medium term, nuclear power is not a cost-effective source of low-carbon electricity for South Australia." Nor did it support removal of restrictions on nuclear fuel cycle activities.

The major recommendation was to set up an international high-level waste storage facility and repository in the state, but the government will not "pursue policy or legislative change" relating to this but would "continue to encourage" discussion on the issue and "remain open to pursuing this opportunity for South Australia" eventually. This would be by restoration of bipartisanship and broad social consent secured through a state-wide referendum as "the only path forward."

In May 2016 a 520 MWe coal-fired power station was closed down and later demolished. It had supplied about one-fifth of state demand. In September 2016 South Australia experienced a state-wide blackout for several hours, prompting re-evaluation of the state’s electricity supply security. Average South Australia wholesale electricity prices increased by 76% between 2015-16 and 2016-17 (from $61.67/MWh to $108.66/MWh).

In mid-2017, total capacity in the state was 5440 MW, consisting of: gas, 2670 MWe (49%); wind, 1700 MWe (31%); rooftop solar PV, 780 MWe; and diesel, 290 MWe.

MIT South Australia report 2018

In October 2018 a white paper from Massachusetts Institute of Technology (MIT) modelled 16 electricity supply scenarios for South Australia. With any carbon emission constraint below 100g/kWh, those with a nuclear component were distinctly cheaper. (SA carbon intensity for 2016-17 was about 290 g/kWh.) It envisaged progressive deployment of small reactors up to 300 MWe each to a total of between 600 and 1500 MWe. The higher figure would support operation of a desalination plant.

Nuclear-powered submarines

In September 2021 the government committed to acquiring a small fleet of nuclear-powered submarines of US or UK provenance for the Navy. The Nuclear Non-Proliferation Treaty (NPT) does not preclude this.

In July 2022 the IAEA was working with Australia over ensuring safeguards in relation to the proposed acquisition of nuclear-powered submarines, a situation Director General Rafael Mariano Grossi described as “complex”.

Earlier background to considering nuclear power

See also Appendix on Australian Research Reactors.

In 1953 the Australian Parliament passed the Atomic Energy Act, which established the Australian Atomic Energy Commission (AAEC). AAEC's functions included advising the Government on nuclear energy matters, and the Commission quickly decided that effective and informed advice could only be provided if there was underlying expertise directly available to it. Hence in 1955 it established a research establishment at Lucas Heights, near Sydney and began assembling a world class team of scientists and engineers. It also began construction of a materials testing reactor, HIFAR, which first achieved criticality and started up on Australia Day, in January 1958.

The AAEC's research program was initially very ambitious and included studies of two different power reactor systems, on the base of substantial multi-disciplinary research in the fields of physics, chemistry, materials science and engineering. Later, recognising Australia's potential as a source of uranium, AAEC also undertook an experimental research program in the enrichment of uranium.

The AAEC also initially convinced the Government that there would be benefits from the construction of a "lead" nuclear power station on Commonwealth land at Jervis Bay, south of Sydney. After competitive bids were obtained, a reshuffle of leadership in the Government led to a loss of interest in the proposal and the project was eventually abandoned in 1972.

In the late 1950s nuclear power was considered for the large new power station at Port Augusta in SA, which was eventually

commissioned in 1963 to burn very low-grade coal from Leigh Creek. In the late 1960s Victoria's State Electricity Commission undertook preliminary studies on building a large nuclear plant on French Island in Westernport. In 1969 the South Australian government proposed a nuclear power plant in SA to supply the eastern states' grid. Then in 1976 the SA government in its submission to the Ranger Uranium Inquiry said nuclear power appeared inevitable for SA, perhaps by 2000.

In 1981, the government's National Energy Advisory Committee presented a report on the administrative and legal issues associated with any domestic nuclear power program. It recommended that "the commonwealth, state and Northern Territory governments should develop with minimum delay a legal framework using complementary legislation as appropriate for licensing and regulating health, safety and environmental and third party liability aspects."

In Australia the possibility of nuclear power is hindered in Victoria and NSW, by legislation enacted by previous governments. In Victoria the Nuclear Activities (Prohibitions) Act 1983 prohibits the construction or operation of any nuclear reactor, and consequential amendments to other Acts reinforce this. In NSW the Uranium Mining and Nuclear Facilities (Prohibitions) Act 1986 is similar. In 2007 the Queensland government enacted the Nuclear Facilities Prohibition Act 2006, which is similar (but allows uranium mining).

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Electricity options

Coal provides the majority of Australia's electricity, and the full picture is given in the Appendix on Australia's Electricity. This also accounts for most of the 140 Mt/yr carbon dioxide emissions from electricity and heat production and uses up about 300 GL/yr of fresh water for evaporative cooling. Preliminary IEA figures show 2020 generation of 258 TWh, 55% from coal and 21% from gas.

Australia is fortunate in having large easily-mined deposits of coal close to the major urban centres in the eastern mainland states. It has been possible to site the major power stations close to those coal deposits and thus eliminate much of the cost and inconvenience of moving large tonnages of a bulky material. Energy losses in electricity transmission are relatively low.

Western and South Australia have relatively less coal but plenty of gas and also lower demand for electricity. More than half of their electricity is derived from burning gas. Development of Tasmania's large hydro-electric resources has put off the day when it needs any large thermal power stations, but hydro potential is now almost fully utilized.

In the next 15 years or so Australia is likely to need to replace the oldest quarter of its thermal generating capacity, simply due to old age. This is at least 8000 MWe, practically all coal-fired. If it were replaced by gas-fired plant, there would be a reduction of about 25-30 million tonnes of CO2 emissions per year. If it were replaced by say six nuclear reactors there would be a reduction of about 50 million tonnes of CO2 emissions per year. Every 22 tonnes of uranium (26 t U3O8) used saves the emission of one million tonnes of CO2 relative to coal.

In other parts of the world as well as Western and South Australia, there was a conspicuous "flight to gas" in the late 1990s while gas prices were low. Generating plant to utilize gas is relatively cheap and quickly built, and at the point of use, gas-fired electricity does cause only half the greenhouse emission of coal. It is clearly an option to utilize more gas for electricity in Australia if low gas prices can be maintained many years ahead.

Moving to gas would be seen by some as a great step forward for the environment. Others would see it as a tragic waste of a valuable and versatile energy resource. Gas can be reticulated to homes and factories to be burned there at much greater efficiency overall.

In January 2007 the Energy Supply Association of Australia (ESAA) completed a study on electricity supply options relative to CO2 emission constraints in meeting projected load in 2030. For a 67% increase in electricity load, greenhouse gas emission targets of 140%, 100% and 70% of 2000 levels were modeled, with three supply options: all credible technologies; without nuclear; and without both nuclear and fossil fuel (with carbon capture and storage). Constraining CO2 emissions would require nuclear contributing 20% of the power, with overall about 30% increase in

costs, hence a need for costing carbon to cover this. ESAA concluded that "the widest possible range of generation technologies will be needed."

Radioactive wastes - domestic

While Australia has no nuclear power producing electricity, it does have well-developed usage of radioisotopes in medicine, research and industry. Many of these isotopes are produced in the research reactor at Lucas Heights, near Sydney, then used at hospitals, industrial sites and laboratories around the country.

Each year Australia produces about 45 cubic metres of radioactive wastes arising from these uses and from the manufacture of the isotopes – about 40 m3 low-level wastes (LLW) and 5 m3 intermediate-level wastes (ILW). These wastes are now stored at over a hundred sites around Australia. This is not considered a suitable long-term strategy.

Since the late 1970s there has been an evolving process of site selection for a national radioactive waste repository for LLW and short-lived ILW. There has also been consideration of the need to locate a secure storage facility for long-lived intermediate-level wastes including those returned to Australia following the reprocessing of used fuel from Lucas Heights. Eventually, disposal options for this will need to be considered also.

Low-level wastes and short-lived intermediate-level wastes will be disposed of in a shallow, engineered repository designed to ensure that radioactive material is contained and allowed to decay safely to background levels. Dry conditions will allow a simpler structure than some overseas repositories. The material will be buried in drums or contained in concrete. The repository will have a secure multi-layer cover at least 5 metres thick, so that it does not add to local background radiation levels at the site.

There is a total of about 3700 cubic metres of low-level waste awaiting proper disposal, though annual arisings are small (the 40 cubic metres would be three truckloads). Over half of the present material is lightly-contaminated soil from CSIRO mineral processing research decades ago (and could conceivably be reclassified, since it is no more radioactive than many natural rocks and sands).

Long-lived intermediate-level (category S) wastes will be stored above ground in an engineered facility designed to hold them secure for an extended period and to shield their radiation until a

geological repository is eventually justified and established, or alternative arrangements made.

There is about 500 cubic metres of category S waste at various locations awaiting disposal, and future annual arisings will be about 5 cubic metres from all sources including states & territories, Commonwealth agencies and from radiopharmaceutical production, plus the returned material from reprocessing spent ANSTO research reactor fuel in Europe. This will be conditioned by vitrification or embedding in cement, and some 26 cubic metres of it is expected by about 2020.

In March 2012 parliament passed the National Radioactive Waste Management Bill 2010 which provides for a national repository for low-level wastes and store for intermediate-level wastes, on land which has been volunteered by its owners, probably at Muckaty Station in the Northern Territory. In April 2016 the government selected one site in South Australia from a shortlist of six, for further consultation.

See further: Radioactive Waste Repository & Store for Australia, as Appendix to this paper.

International waste repository proposal

In May 2016 the South Australian Royal Commission into the Nuclear Fuel Cycle reported. A major recommendation was that a facility for the interim storage and subsequent disposal of international used nuclear fuel and intermediate-level waste should be established. It found that the state "has the necessary attributes and capabilities to develop a world-class waste disposal facility, and to do so safely." Interim above-ground storage for used fuel inside heavily engineered, purpose-built dry casks would be followed by deep geological disposal, the repository design based on that under construction at Olkiluoto in Finland at 400-450 metre depth. In the analysis, the geological disposal facility for used fuel is notionally co-located with an intermediate-level waste facility, where those packages are placed in medium-depth vaults of 50-250 metres. The report said that integrated facilities with capacity to store and dispose of used fuel would be viable.

Based on a "cautious and conservative approach", from assessments of used fuel inventories and potential global interest the commission determined that such a facility could generate more than A$ 100 billion in income in excess of expenditure (including a reserve fund of A$ 32 billion for facility closure and ongoing

monitoring) over the 120-year life of the project. The quantities assumed are about one-eighth of the used fuel (138,000 t) and intermediate-level wastes, commencing in year 11 with 3000 t/yr used fuel to interim storage.

The World Nuclear Association said that the report had "fundamentally changed the nature of the global nuclear waste discourse", and a multinational waste facility based in South Australia would provide a welcome option for countries operating nuclear facilities today. It would be a “viable alternative" to national projects. Such a large multinational waste storage facility would be a world first and should offer advantages in terms of siting and economics when compared to smaller national approaches.

A fuller account of the proposal is in the information paper on International Waste Disposal Concepts.

Research & development, isotope production

The High Flux Australian Reactor (HIFAR) operated at Lucas Heights near Sydney from 1958 to 2007, and was for many years the only operating nuclear reactor in Australia. It was used for materials research, to produce radioactive materials for medicine and industry and to irradiate silicon for the high performance computer industry. It was a 10 MW unit which had the highest level of availability of any research reactor in the world. It was at the heart of almost all the research activities of the Australian Nuclear Science and Technology Organisation (ANSTO) and supported those of several other organisations on the same site.

In 2006 HIFAR was replaced by a new research reactor, known as OPAL (Open Pool Australian Light-water reactor), a modern 20 MW neutron source. It achieves over 300 operational days per year, in the top league of the world's 240 research reactors. OPAL uses low-enriched fuel and for Mo-99 production it irradiates low-enriched targets which are then processed to recover the Mo-99.

In December 2016 the draft National Infrastructure Research Roadmap was released, setting out priorities for the decade ahead. Research infrastructure such as the OPAL reactor and the Australian Synchrotron in Melbourne are recognized in the document as "a category of research infrastructure that is of such scale that the national interest is best served by landmark facilities," which rely on government investment.

In September 2012 ANSTO announced a A$ 168 million expansion of its Sydney facilities, principally for molybdenum-99 production, the source of technetium-99 which is widely used in nuclear medicine for diagnosis. Current world demand is about 45 million doses (23,000 six-day TBq/yr) per year, and the new plant will be capable of meeting about one-quarter of this from 2016 at a time when the main plants in Canada and Europe are set to close. The new plant will more fully utilize the OPAL reactor's capacity. The investment also covers building an industrial-scale plant for Synroc waste form to immobilize the intermediate-level wastes from Mo-99 production. ANSTO hopes that the Synroc technology "will become the benchmark for waste treatment in the production of Mo-99 radiopharmaceuticals."

See further: Australian Research Reactors, as Appendix to this paper, and Synroc Wasteform appendix to the information paper on *Treatment and Conditioning of Nuclear Wastes*.

Non-proliferation

Australia's uranium is sold strictly for electrical power generation only, and international safeguards are in place to ensure this. Nuclear equipment and services are also for peaceful uses only. The Australian Safeguards & Non-proliferation Office (ASNO) assists the International Atomic Energy Agency (IAEA) by allowing access to Australian nuclear facilities and arranging for the installation of safeguards equipment as required. It reports regularly to the IAEA on nuclear material held in Australia.

Australia is a party to the Nuclear Non-Proliferation Treaty (NPT) as a non-nuclear weapons state. Its safeguards agreement under the NPT came into force in 1974 and the Additional Protocol in relation to this came into force in 1997. A bilateral safeguards agreement is required with each customer nation as a precondition of trade, placing additional requirements on them beyond those of the NPT and the IAEA. Australia has 25 bilateral safeguards agreements covering over 40 countries (the Euratom agreement covering all 28 countries in the EU, giving some overlap). Australia is also a member of the Nuclear Suppliers Group.

Australia has in place an accounting system that follows uranium from the time it is produced and packed for export, to the time it is reprocessed or stored as nuclear waste, anywhere in the world. It also includes plutonium which is in the spent fuel or recovered from it. All documentation relating to Australian obligated nuclear

material (AONM) is carefully monitored so that any apparent discrepancies can be taken up with the country concerned. There have been no unreconciled differences in accounting for AONM. This system operates in addition to safeguards applied by the IAEA which keep track of the movement of nuclear material through overseas facilities and verify inventories.

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Appendix

A brief history of Australian uranium mining

The existence of uranium deposits in Australia has been known since the 1890s. Some uranium ores were mined in the 1930s at Radium Hill and Mount Painter, South Australia, to recover minute amounts of radium for medical purposes. Some uranium was also recovered and used as a bright yellow pigment in glass and ceramics.

Following requests from the British and United States governments, systematic exploration for uranium began in 1944. In 1948 the Commonwealth Government offered tax-free rewards for the discovery of uranium orebodies. As a result, uranium was discovered by prospectors at Rum Jungle in 1949, and in the South Alligator River region (1953) of the Northern Territory, then at Mary Kathleen (1954) and Westmoreland (1956) in north west Queensland.

In 1952 a decision was taken to mine Rum Jungle, NT and it opened in 1954 as a Commonwealth Government enterprise. Radium Hill, SA was reopened in 1954 as a uranium mine. Mining began at Mary Kathleen, Qld in 1958 and in the South Alligator region, NT in 1959. Production at most mines ceased by 1964 and Rum Jungle closed in 1971, either when ore reserves were exhausted or contracts were filled. Sales of some 7730 tonnes of uranium from these operations were to supply material primarily intended for USA and UK weapons programs at that time. However much of it was used in civil power production.

The development of nuclear power stimulated a second wave of exploration activity in the late 1960s. In the Northern Territory, Ranger was discovered in 1969, Nabarlek and Koongarra in 1970, and Jabiluka in 1971. New sales contracts (for electric power generation) were made by Mary Kathleen Uranium Ltd., Queensland Mines Ltd. (for Nabarlek), and Ranger Uranium Mines Pty. Ltd., in the years 1970-72.

Successive governments (both Liberal Coalition and Labor) approved these, and Mary Kathleen began recommissioning its mine and mill in 1974. Consideration by the Commonwealth Government of additional sales contracts was deferred pending the findings of the Ranger Uranium Environmental Inquiry, and its decision in the light of these. Mary Kathleen recommenced

production of uranium oxide in 1976, after the Commonwealth Government had taken up a 42% share of the company.

The Commonwealth Government announced in 1977 that new uranium mining was to proceed, commencing with the Ranger project in the Northern Territory. In 1979 it decided to sell its interest in Ranger, and as a result Energy Resources of Australia Ltd was set up to own and operate the mine. The mine opened in 1981, producing 2800 t/yr of uranium, sold to utilities in several countries. Production over three years to mid 2002 averaged 3533 t/yr of uranium.

In 1980, Queensland Mines opened Nabarlek in the same region of Northern Territory. The orebody was mined out in one dry season and the ore stockpiled for treatment from 1980. A total of 10,858 tonnes of uranium oxide were produced and sold to Japan, Finland and France, over 1981-88. The mine site is now rehabilitated.

At the end of 1982 Mary Kathleen in Queensland had depleted its ore and finally closed down after 4802 tonnes of uranium oxide had been produced in its second phase of operation. This then became the site of Australia's first major rehabilitation project on a uranium mine site, which was completed at the end of 1985. A Rum Jungle Rehabilitation project also took place in the 1980s.

Australian Labor Party (ALP) policy on uranium mining has varied over four decades. The 1971 Platform, on which the Whitlam Government was elected in 1972, committed the party to working towards the establishment of a domestic uranium enrichment and nuclear power sector. But after losing government in 1975, pressure grew in the Labor Party for a strong stance against uranium mining and export, as a counterpoint to Liberal Coalition policies to expedite uranium mining and export. An anti-nuclear movement gained strength and campaigned to end Australian uranium mining.

The 1977 ALP National Conference adopted a new policy. Community concerns with the threat of nuclear war were to be allayed by ending uranium mining and ceasing Australia's contribution to the nuclear fuel cycle. The change committed a future Labor government to declare a moratorium on uranium mining and treatment and to repudiating any commitments to mining, processing or export made by a non-Labor government. The policy made a strong statement and was seen to provide moral leadership.

By the time of the 1982 ALP National Conference, many in the Labor Party were troubled about how an incoming Labor

Government would implement the party's moratorium policy. There was concern that the repudiation of contracts would raise issues of sovereign risk and would expose a Labor government to compensation liabilities. An amendment to the ALP Platform in 1982 sought a compromise between the positions of those who wanted to shut the industry down and those who felt that doing so was neither possible nor in the national interest. It committed Labor to a policy on uranium mining which was a classic political compromise, the core of which endured as Labor policy for 25 years. The policy was designed to prevent new uranium mines; limit Australia's uranium production with a view to the eventual phasing out of mining altogether; and provide moral leadership in ending the nuclear industry.

However, in a concession to South Australia, it also said that a Labor government would "consider applications for the export of uranium mined incidentally to the mining of other minerals on a case by case basis". This was the Roxby Downs amendment, which would allow export of uranium from Olympic Dam - a major copper and uranium deposit. So the 1982 anti-uranium policy actually authorized the development of the world's largest uranium mine!

In the 1983 federal election the ALP won office. The 1984 ALP National Conference then dropped the language of moratorium, repudiation of contracts and phase-out from the Platform. For the first time the three-mines-policy was delineated by naming Nabarlek, Ranger and Roxby Downs (Olympic Dam) as the only projects from which exports would be permitted. Provisional approvals for marketing from other prospective uranium mines were cancelled.

The naming of specific mines was later deleted from the Labor Platform in the light of the fact that Nabarlek ceased production in 1988 and under a (conservative) Coalition government Beverley started up in 2000. The ALP policy then only allowed exports from existing mines and prevented the establishment of new ones. This endured as a "no new mines" policy through a change of government in 1996 until 2007, when it was abandoned as ineffective and likely to be electorally negative due to changed public opinion arising from global warming concerns. Opposition to uranium mining was then left to state ALP branches and governments. This continued in WA and Queensland until changes of government in 2008 and 2012 respectively. (NSW and Victoria have legislation banning uranium exploration and mining, which has not been repealed by conservative governments.) A fourth mine, Honeymoon, started up in South Australia in 2011.

Adapted from Senator Chris Evans speech 23/3/07 to Labor Business Roundtable, Perth.

During 1988 the Olympic Dam project commenced operations. This is a large underground mine at Roxby Downs, South Australia, producing copper, with uranium and gold as by-products. Annual production of uranium started at some 1300 tonnes, with sales to Sweden, UK, South Korea and Japan. After a A$ 1.9 billion expansion project, production increased to over 4000 tonnes uranium per year by mid 2001. In 2005 it was taken over by BHP Billiton.

Beverley, the first Australian in situ leach (ISL) mine started up in 2001 and closed early in 2014. Another ISL mine, Honeymoon, came on line in 2011 and closed in 2013. Production from Four Mile started in 2014, using a satellite plant to capture the uranium from ISL and the Beverley plant for final product recovery. Beverley and Honeymoon may resume production with increased uranium prices.

Both Ranger and Nabarlek mines are on aboriginal land in the Alligator Rivers region of the Northern Territory, close to the Kakadu National Park. In fact the Ranger and two other leases are surrounded by the National Park but were deliberately excluded from it when the park was established. Ranger is served by the township of Jabiru, constructed largely for that purpose. Nabarlek employees were based in Darwin and commuted by air.

See also: Former Australian Uranium Mines Appendix.

Appendices

Australia's Uranium Mines
Australia's U deposits and Prospective Mines
Australia's former uranium mines
Australia's Electricity
Australian Research Reactors
Radioactive waste repository & store for Australia